MSX and SPECTRAVIDEO

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COMPUTER FORUM
Magazine

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EDITORIAL

Another year nearly gone and Christmas is with us again. I don't know what you want Santa to bring you. I would like a stocking full of press releases about MSX computers, peripherals and software being released in Australia.

Some readers have written requests for more software reviews for Spectravideo. Computer Forum is produced within four limits. These are a balance of content between MSX and Spectravideo and within both areas a smattering for people who range from novice to hacker.

We are able to obtain quite a range of software for review, but try to restrict our reviews to products which we know, or hope, are generally available to our readers. There is not a great deal of point in reviewing the latest MSX2 releases or a Spectravideo title we know to have been out of stock for the past six months. The majority of the time taken to publish Computer Forum is devoured in obtaining items for review and researching readers' queries.

This is not a complaint, as we have received some tremendous support from the distributors and retailers contacted. It would be nice though to have some review samples or press releases arrive without having to spend $100 in phone calls.

Ah well, Santa. If you can't manage that, then could you see your way clear to a bottle of scotch to drown the sorrows.

Merry Krishmush ...

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PRODUCT NEWS ...

EXTRA MSX MEMORY

A 32K RAM cartridge has been produced by Forrest Data Services for use with the Yamaha CX5M and Pioneer PX-7 computers. This brings them up to the full 64K RAM which is necessary for running certain software. Manufactured in Australia and priced at $135 this is a cheaper alternative to the existing units which are also difficult to obtain.

LASER DID NOT MAKE IT

In the first issue of Computer Forum it was reported that the Laser MSX was due for release in Australia this year. The latest information is that it has been shelved.

Mr Fung of Video Technology advised that, due to sourcing components from Japan and the current strength of the Yen, the Laser will not be manufactured in either MSX1 or MSX2 in the foreseeable future.

Such a shame, as this looked like being the one to buy due to the ability of the MSX1 to accept an MSX 2 adaptor.

THE INS AND OUTS OF MSX

The Mitsubishi MSX is out. The National and JVC never came in. Sony have several MSX units to quit before they consider MSX2. Toshiba are still maintaining that they will have their MSX2 here in the new year. Sanyo are still trying to establish their market. The four major software distributors — Ozi-Soft, Imagineering, ISD and Melbourne House are no longer listing MSX software on their price lists. Toshiba and Sony still have some stock of their own titles — at present.

SPECTRAVIDEO — WHICH WAY?

Spectravideo kicked off the area of computing in which we are all now involved. Where are they heading now?

Spectravideo started off with the SV300 range of computers which they thought were MSX standard. When Microsoft changed the MSX standard and offered it to other manufacturers, SVI brought out their SV I700 range of MSX computers. The latest of these was the SVI 738 X'press.

Recent communications with Spectravideo indicate that the 300 and 700 range are no longer in production. The 738 is still available but only manufactured to order (100 pieces). An enquiry from this magazine about the possibility of the 738 being an MSX2 with MSX1 chip package brought the reply that SVI 'have no intention of following MSX2. Our future development will be based on 8088 microprocessor design'.

For the uninitiated, 8088 micro technology is used in IBM compatibles. In fact, Spectravideo have been producing IBM compatibles for some time. The SVI 256 and SVI 640 range would appear to be the Bondwell IBM compatible.

The latest Spectravideo retains two links with the last of the MSX machines. It is called the X'press 16 and is IBM compatible with 256K VRAM for enhanced hi-res graphics. The Advanced Video Display Processor (AVDP) gives 256 x 212 (256 colour) resolution and 512 x 212 (16 colour) resolution and 32 multi colour sprites. The Sound chip has 8 octaves and 3 channels. (Sounds familiar.)

The other link with MSX comes from the SVI 811 MSX Games Adaptor. Our first impression was that this was the machine to buy — IBM compatibility with the excitement of MSX capability. Unfortunately the adaptor will only run MSX cartridge software, not tape or disk. The adaptor will not work with other IBM clones. The X'press 16 comes with MS-DOS and enhanced GW-Basic (which is a cousin of MSX Basic). So, if you are thinking of buying a clone, but want to retain a link with MSX then ...

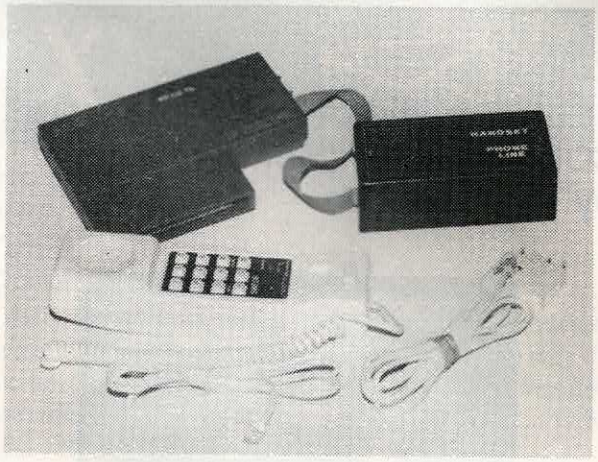
MSX MODEM

We have been checking out the Interlink Modem. The unit consists of the Modem/RS232 cartridge, telephone interface, telephone and cables and is very simple to set up. The cartridge fits in the cartridge slot, the telephone plugs into the interface which then plugs into the phone socket.

The Modem has built-in firmware to handle Viatel and normal communications using 300 baud Full Duplex and 1200/75 split baud rate. Both CCITT and BELL standard are used and the handset has last number re-dial (which we found handy during busy periods on Viatel). To access Viatel, type CALL VIDEOTEX and set one switch to 'originate' and the other to 'phone'. Dial the viatel number and when the carrier signal is heard (a high pitched whistle), set the switch to 'modem'. At this point the introductory Viatel screen is displayed and you are asked for a password. If you do not have a Viatel account you will not get past this point.

On entering Viatel, the main index is presented. Selecting from this index will usually present you with another, more detailed index. Any screen presented can be dumped to your printer for later use.

The Interlink Viatel Modem is suitable for all MSX computers which do not have an RS232 interface. This means it is not usable with the SVI 738. We do not know if it will work with the Yamaha CX5M or Pioneer PX-7.



This system opens up a whole new area of computing and our thanks to Telecom for providing us with a temporary Viatel account to evaluate this package. The Interlink MSX Modem package is made in Australia and sells for $399.

STOP PRESS: The Interlink Modem has been tested with the Yamaha CX5M and Pioneer PX-7 and worked well with both systems.

Software NEWS & REVIEWS

MEGABASE (MSX Disk) by FDS Software

This is a file card data base for use with MSX Disk systems. Files can be set up under user created categories with the file containing a title and data, the size of which can be set by the user from 1 to 15 lines of 40 characters.

Another essential feature is the ability to use headings for each line of data onto each file category. This is very useful for keeping track of club members or addresses. All the usual search file modes are included and a file is found at surprising speed.

The program is compatible with most printers including MSX plotters and Epson compatible printers.

Price: $29.95.

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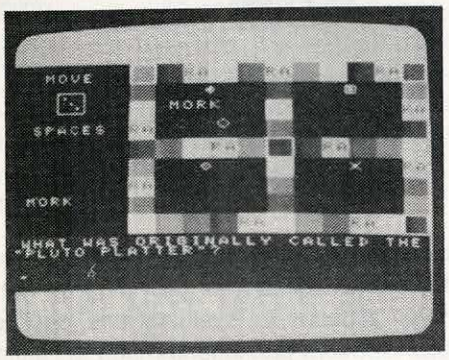
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Software NEWS & REVIEWS

SPECTRA-TRIVIA (SV328 Cassette)
COMPU-TRIVIA (MSX Cassette)
by FDS Software



The famous board game now on computer. Available in SV and MSX format cassette, these are trivia quiz games for up to four players. The computer rolls the dice for each player who then moves his playing piece with the joystick. The winner is the first to correctly answer questions on each of the outer corner squares and then the centre red square.

Each version has the facility to create your own question and answer file. Unfortunately the program is too large to operate on an unexpanded SV318. Who says computer games are all space shot outs? This is challenging family entertainment.

Price: $19.95.

BONE OF CONTENTION (SV318/328 Tape) by Spectravideo

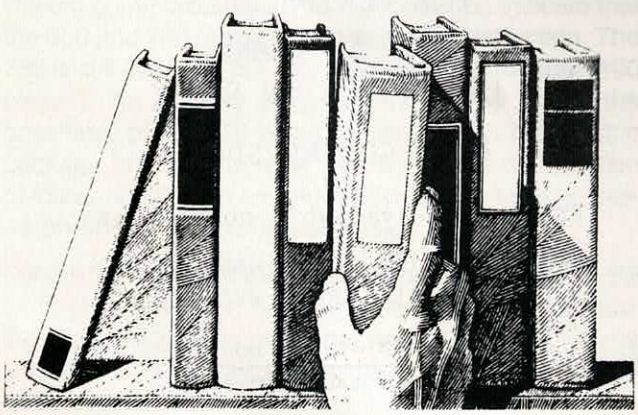


This is an educational maths game for two players. Two dogs are playing tug-o-war with a bone. To get your dog pulling in the right direction, you must correctly answer the maths question set by your opponent and enter one for him to solve.

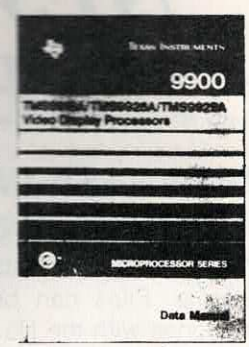
This is quite a good test of maths under pressure. The questions are usually a formula containing addition, subtraction, multiplication and division. The only criticism is the use of the colon as a divide symbol.

Price: $12.00.

BOOKS • BOOKS



9900 DATA MANUAL



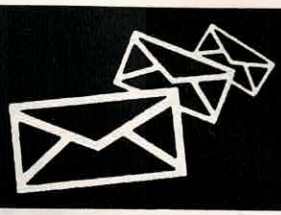
This is the technical manual for the 9929 Video Display Processor chip used in your SV or MSX. Although written in fairly technical terms, it does explain the different screen modes, the sprite planes and registers used to generate the display.

The relationship between the different tables used in graphics modes is also covered and well illustrated with diagrams. An appendix shows the sprite patterns used to display text. Interfacing data and video RAM control are also covered which is useful to the more technical reader.

Our copy came from VSI Electronics. They should have an office in your state.

Publisher: Texas Instruments
106 pages
Price $10.00

MAILBOX



This page is for your comments and queries. If you have a specific problem or have solved one, we would like to hear about it.

If you require an answer before we publish it, please enclose a stamped self-addressed envelope.

TAPE PROBLEMS

I also have the same problems as Colin Johnstone (Volume 1 No 5). Like Colin I own the Sony Hit Bit and could not load programs at the fast (2400 baud) rate. However I followed your instructions robotically on loading and Bingo! My computer loaded 2400 for the first time. But just ten minutes later I tried saving and loading a program at 2400 and it just didn't work. I noticed I was using a different brand of tape, so tried the original one and that didn't work. Trying again later, I found that it would only work about 20% of the time and only with the first brand of tape. The recorder is the "proper" Sony Bitcorder. Is there a flaw in the great Hit Bit?

Paul Rode
Curlewis, NSW

As mentioned in issue 5, the 2400 baud rate can be very touchy. Most CLOAD problems are the result of a bad CSAVE. If the program only partly loads, then make a note of the counter reading where it crashes. If this is the same each time, then there may be a flaw in the tape at that point or a power 'blip' was recorded at the time of CSAVEing.

If the program is not being found at all then this can be caused by incorrect volume settings (on recorders with that facility) or possibly dirty heads on the recorder. If you are using very high quality tapes such as Chrome Dioxide or Ferrochrome then that may be the problem. The recording signal is designed for Low Noise tapes. These are usually the cheapest, but steer clear of the '2 for $1' supermarket types. It took FDS Software several months to find a reliable tape for mastering and producing cassette programs.

MORE MEMORY

My knowledge of computing is far less than my enthusiasm. I have the SV807 64K RAM card added to my expander and find it quite handy with the SWITCH command. But I desperately need to add it on to my built-in memory capacity so giving me a chance to run larger programs and files. Is there any way I can enlarge my memory capacity either in CP/M or SV Basic, preferably both?

Tony Jaffrey
Clontarf, Queensland

I think we all have more enthusiasm than knowledge when it comes to computers. The only effective way that we have seen to utilise more memory in Basic has been with MEMORY SHIFT which is produced by TIL, PO Box 138, Gnowangerup 6335.

CP/M 3.0, which is used on some other computers, makes use of 128K of RAM. We do not know of anyone who has this available for the Spectravideo.

PEEKING PROBLEMS

As a very raw but ancient (60 + +) novice in the computer programming field, I feel I must commend you on a magazine that is informative and instructive without being too technical. In fact, that takes time out to explain much of the technical (and to me strange) language involved.

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I was very impressed by the straightforward comment on de-bugging problems found on page 29, issue No 6. Whilst I can appreciate the frustration of a program that 'appears' to be wrong, my short experience has taught me that patience IS a virtue, and one must constantly re-check for typing errors.

Peeking and poking (issue 6, pages 12-14) I get totally different results to those shown. In the one line program (page 13) using the format for SV318, the resulting display is nothing like that shown.

The 'flag' value on page 14 — the SV format gives no result, the MSX format gives the anticipated result — a cursor that varies between full and half height.

R Mahoney
Port Kembla, NSW

Thank you for the kind words and you are right. There are different versions of SV ROM which can expect different areas of the system RAM to contain the information we provide. Usually the locations given are within a few bytes of each other, and checking around that area will usually disclose the required information. Try PEEKing 64896 to test for the "A" key.

Likewise, the addresses given on pages 13 and 14 are correct for our (fizz, crackle) old MKI 318. Some of these locations were found by trial and error — print out the contents of the entire system RAM, change some things with the keyboard and print it out again. We end up with some interesting data and a couple of boxes of used printer paper.

DISK AND OTHER QUERIES

I am using a SV328 with two disk drives and 80 column card and would like to know how to make the computer boot up to the 80 column card. Another problem is that SVRFMT will not operate while using the 80 column card. Both of these would need an alteration to the programs on the disk. Also COMMON appears to be a reserved word. How is this used? CHAIN is listed as a reserved word but does not appear to be so as the computer accepts it as a variable name. Finally, is there an assembler available to run under disk basic?

R J Thomson
Nambour, Queensland

The 8O column boot was covered in our first issue, and involves adding a line containing WIDTH 80 to the IPL program. There does appear to be a clash between SVFRMT and the 80 column card with only the first two tracks formatting. Has anybody fixed this? Much of the SV manuals were borrowed from existing text covering MBASIC and BASICA and contain some commands which are not supported in SV Basic — such as CHAIN and COMMON. We are not aware of a commercial disk based assembler for the SV. You might get some joy by contacting the user groups listed in Computer Forum.

KEY PROBLEM

My problem is that part of my keyboard will not work. Every time I press this key it beeps, but nothing comes up on the screen. The key is the one next to the DEL/CUT key on my SVI 728. Can you help me?

Jason Ward
Taree, NSW

The key you describe on your MSX is known as the DEAD key. This is used to produce accents on certain letters. You have probably seen these on European labels etc. For example, the two dots above the letter "o" etc. Try holding this key down and pressing the lower case "a" key.

PARALLEL TO SERIAL

We have had a request from Mr K Creek of Glen Waverley for information about using a serial printer with a SV318 and RS232 card. The problem is that LPRINT and LLIST assume a parallel printer, and he was wondering if there is any way of redirecting these commands to the RS232 card.

The simplest? ways we know of doing this are to write your own driver routine and call that routine instead of using LPRINT or LLIST, or to construct a parallel to serial converter to attach to a parallel printer interface. Has anyone overcome this problem?

MSX SOFTWARE

The cover of Volume 1 No 5 showed a brochure of sorts from what seems to be a Japanese MSX catalogue. I am most interested in "ZORNI" by Toshiba, yet their distribution centre in North Ryde know nothing about it and suggested I contact Toshiba (Japan) 1-1-1 Shibaura, Minato-Ku, Tokyo, Japan. Could you publish the address (of the catalogue) supplier, for I am sure many MSX users are dying to get hold of top material like that.

Talking to a contact from Sony in Chatswood, MSX is the biggest thing ever in Japan, but here? Well, he is not impressed with sales of Hit Bits despite demos of games like PAYLOAD and SCION. More inside information from Sony confirmed that they will be releasing a COBOL compiler cartridge for MSX/MSX2.

Roger Stockburger
Richmond, NSW

The cover of number 5 was, in fact, extracted from the MSX Soft Catalogue supplied by ASCII Corporation. This is a 160 page catalogue of MSX and MSX2 software. Unfortunately it is all in Japanese (addresses too).

One reason MSX is so big in Japan is the amount of advertising. The Japanese MSX magazine is 218 pages with Sony taking two to three double-page colour ads each month. When was the last time you even saw an ad for any MSX in Australia?

COMPUTER FORUM Magazine
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BASIC TUTORIAL — Part 8

RANDOM NUMBERS

The RND command

This command is used when you want the computer to select numbers at random. These might be for a game or to select values for a maths practice program. It takes the form

```
RND(x)
```

where x can be either 0, 1 or -1. Its purpose is to generate a random number between 0 and 1. This is fine if you want a decimal number but you will most likely want a whole number or integer. Therefore, we must multiply the random number by nine and convert it to integer to obtain a whole number between 0 and 9.

For example

```
10 PRINT INT(RND(1)*9)
```

RUN this program a few times. You will notice the same number is given each time - not very random. Try adding line 20 as follows

```
10 PRINT INT(RND(1)*9)
20 PRINT INT(RND(1)*9)
```

Now run the program and you will see two random numbers which are different but, each time you run it, the numbers will be the same and in the same sequence. This is because RND is not a true RANDOM command. This is because the computer generates a list of random numbers and the RND command selects the next one on the list each time it executes.

To generate truly random numbers, we can use the negative value of the internal timer. The expression would now be RND(-TIME). Now try this

```
10 PRINT INT(RND(-TIME)*9)
20 PRINT INT(RND(-TIME)*9)
```

DEFINING VARIABLES

Variable types were covered in Part 1 of this tutorial. It was explained that the $ sign behind a variable name indicated a string variable and a % sign indicated an integer variable. Other signs used are the ! for single precision and # for double precision variables.

What does this mean? An integer is a whole number - 1 and 2556 would be integers. A single precision number is one that has up to 6 decimal places of precision - 1.234567 and a double precision number can have up to 14 decimal places.

```
Example - A%=123 (integer)
          A!=123.456789 (single precision)
          A#=123.45678912345678 (double precision)
```

The SV318, SV328 and all MSX computers use single precision as a default. That is, if you do not specify the precision, all mathematics are carried out to six decimal places. That is why, when you include a number larger than 32767 in a program line, it will have ! appended to it when you list the program.

```
For example      10 X=40000 : PRINT X
will list as     10 X=40000! : PRINT X
```

In the majority of games programs, numeric variables only need to be integer format. Therefore, defining them as integer will speed up the program. Try these two programs for speed comparisons -

```
10 FOR I=1 TO 20000 : NEXT I          10 FOR I%=1 TO 20000 : NEXT I%
```

The first example will take approximately 53 seconds to run and the second only 31 seconds.

Apart from using the %, !, # and $ symbols to specify variable type, the commands DEFINT, DEFSNG, DEFDBL and DEFSTR can perform the same function for the whole of the program. For example, using DEFSTR A in this program means that wherever the variable A appears in the program (without a trailing variable specifier) it will be taken as being the same as A$.

```
10 DEFSTR A
20 A="ABC"
30 A=A+"DEF"
40 A%=123
```

This means that if, for most of the program, a variable is to be a certain type then that type can be specified at the start of the program but still allows the variable to be a different type if specified at time of using.

JOYSTICK COMMANDS

The STICK command

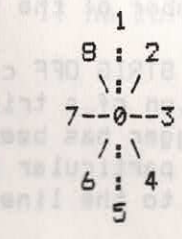
This command is used to read the value of a joystick. The value relates to the direction in which the joystick is moved. It takes the form

STICK (x)

where x can be a number between 0 and 2. If x=0 then the keyboard cursor controls will be read, while x=1 or x=2 reads the joystick plugged into ports 1 and 2 respectively.

Example 10 PRINT STICK(1) : GOTO 10

will print a value according to the direction the stick in port No 1 is moved in relation to the diagram to the right -



If you want to check for a direction on any of the three joysticks, use the formula

S=(STICK(0)+STICK(1)+STICK(2))

With this, you do not need to specify which joystick is to be used.

The STRIG command

This command is used to check for a trigger being pressed. It takes the form -

STRIG(x)

For SV318/328, x can have a value of 0 to 2 which corresponds with the joystick to be read. MSX computers can use dual trigger joysticks and x can have a value between 0 and 4 with 0 being the spacebar, 1 or 3 being the triggers on joystick 1 and 2 or 4 for the triggers on joystick 2.

Example

```
10 X=STRIG(0)
20 Y=STRIG(1)
30 PRINT X,Y
40 GOTO 20
```

This program will print two columns of zeroes down the screen. When the spacebar is pressed the left figure will change to -1 and when the trigger on joystick 1 is pressed the right figure will change to -1.

The STRIG ON/OFF/STOP command

These commands are used to switch on or off the detection of a trigger being pressed. They must be used for the following ON STRIG command to work. They take the forms

```
STRIG(x) ON
STRIG(x) OFF
STRIG(x) STOP
```

where x is the number of the trigger concerned.

The STRIG ON and STRIG OFF commands are self explanatory. They turn on or off the detection of a trigger press. The STRIG STOP command is used to check if a trigger has been pressed prior to a STRIG ON command. Then as soon as that particular trigger is turned on, (STRIG (x) ON) the program will jump to the line specified by the ON STRIG command.

The ON STRIG command

This command is used to branch off to a routine whenever a trigger is pressed. It takes the form

```
ON STRIG(x) GOTO
ON STRIG(x) GOSUB
```

where x is the number of the trigger being checked. In the first example, when a trigger is pressed, processing will continue at the line number following the GOTO command. In the second example, processing will continue with the subroutine following the GOSUB command.

For example

```
10 ON STRIG GOSUB 100,200
20 STRIG(0)ON:STRIG(1)ON
30 GOTO 30
 .
 .
100 PRINT"SPACEBAR PRESSED"
110 RETURN
200 PRINT"EXTERNAL TRIGGER PRESSED"
210 RETURN
```

GRAPHICS COMMANDS

The PRESET command

This command is used to set or reset a point (pixel) on the graphics screen and takes the form -

```
PRESET (a,b),c
```

where a is the horizontal point on the screen, b is the vertical point on the screen and c is the colour to set the point. The co-ordinates a and b can have a value from -32768 to 32767 but only 0 to 255 for a and 0 to 191 for b will display on the screen. If no colour is specified, then the current background colour is used.

Example

```
10 SCREEN 1 : REM MSX USE SCREEN 2
20 PRESET(20,20),1 : GOSUB 80
30 PRESET(21,24),15: GOSUB 80
40 PRESET(22,28),1 : GOSUB 80
50 PRESET(23,32),15: GOSUB 80
60 PRESET(24,36),1 : GOSUB 80
70 GOTO 70
80 FOR T=1 TO 100 : NEXT T
90 RETURN
```

The effect is different in low resolution Screen 2 (MSX Screen 3). Change line 10 to SCREEN 2 for SV or SCREEN 3 for MSX. Remember that the high resolution is 0-255 across and 0-191 down. The low resolution screen uses the same co-ordinates but actually uses a STEP 4 method of displaying graphics. Therefore 20 across is the same as 21, 22 or 23 across.

The PSET command

This command is also used to set or reset points on the graphics screen and takes the form -

```
PSET (a,b),c
```

where a is the horizontal point on the screen, b is the vertical point on the screen and c is the colour to set the point. The co-ordinates a and b can have a value from -32768 to 32767 but only 0 to 255 for a and 0 to 191 for b will display on the screen. If no colour is specified, then the current foreground colour is used.

The POINT command

This command is used to check the colour of a point on the graphics screen and takes the form -

```
X = POINT (a,b)
```

where a is the horizontal point, b is the vertical point to be checked and X will contain the number relating to the colour of the point being checked.

The co-ordinates a and b can have a value from -32768 to 32767 but only 0 to 255 for a and 0 to 191 for b will check the screen. All other values will give -1 as the returned value.

Example

SV Version

```
10 SCREEN 1
20 FOR C=1 TO 15
30 LINE(100,100)-(110,110),C,BF
40 LINE(50,10)-(0,0),4,BF
50 PRINT POINT(105,105)
60 FOR T=1 TO 100 : NEXT T
70 NEXT C
```

MSX Version

```
10 SCREEN 2
20 OPEN"GRP:"AS#1
30 FOR C=1 TO 15
40 LINE(100,100)-(110,110),C,BF
50 LINE(50,10)-(0,0),4,BF
60 PRINT#1,POINT(105,105)
70 FOR T=1 TO 100 : NEXT T
80 NEXT C
```

SPRITES

REFINING SPRITE COLLISIONS

The ON SPRITE command, explained last month, checks for two or more Sprites attempting to occupy the same location on the screen. Sometimes, with more complex graphics, you will use two or more overlaid sprites to depict an image. This will continually trigger the ON SPRITE command and make it unusable.

To check for sprite collision in this case, we must continually compare the screen locations of the Sprites in question. The difference being that the ON SPRITE command is more efficient because it checks sprite locations before it carries out any command anywhere in the program.

We can only compare screen locations, whenever our checking routine is executed.

Example

```
10 SCREEN 1,2 : REM MSX USE SCREEN 2,2
20 A$=STRING$(8,255)
30 SPRITE$(1)=A$
40 SPRITE$(2)=A$
50 PUTSPRITE1,(200,96),1,1
60 X=X+1
70 IF X<200 THEN PUTSPRITE2,(X,96),15,2 : GOTO 60
80 LINE(190,90)-(220,120),6,BF
90 GOTO 90
```

THE BASICS OF DISK BASIC

Disk Basic is an extension of the Microsoft Basic used by SV318/328 and MSX computers. The extra commands are automatically loaded from the disk and held in RAM. That is why a disk system has slightly less usable RAM than a cassette system. If you own, or are comtemplating buying disk drives, this section will give you an introduction to the commands used by Disk Basic.

The FILES command

This is one of the first disk commands that you will learn. It is used to discover which files (or programs) are contained on the disk. It takes the forms

SPECTRAVIDEO DISK BASIC

```
FILES     Displays all files on drive #1
FILES 1   Displays all files on drive #1
FILES 2   Displays all files on drive #2
```

MSX DISK BASIC

```
FILES            Displays all files on drive A
FILES "A:"       Displays all files on drive A
FILES "B:"       Displays all files on drive B
FILES "*.*"      Displays all files on drive A
FILES "*.BAS"    Displays all files on drive A which have .BAS suffix.
```

All file names on an MSX disk are converted to upper case and it does not matter, when loading a file, whether the name is entered in upper or lower case. Spectravideo Disk Basic is different. Upper and lower case names are stored without conversion and file name XXXX is different from file name xxxx. It also cannot display only files with a certain suffix.

The LFILES command

This command is the same as the FILES command, except the information is sent to the printer instead of the screen. If a printer is not connected then the computer will freeze and have to be reset or turned off and on.

The KILL command

USE WITH CAUTION!

YOU CANNOT UN-KILL A FILE!

This command is used to remove a file (or program) from the disk. Actually, the file is still on the disk, but the first character of the filename in the directory track is changed to a space. It takes the forms -

SPECTRAVIDEO

KILL"d:xxxxxx"

where d can be 1 or 2 to indicate the drive number and xxxxxx is the name of the file to be deleted.

ONLY KILL FILES FOR WHICH YOU KNOW YOU HAVE ANOTHER COPY. Then if you make a mistake, you still have a copy, and if all is well then you can KILL the backup copy.

MSX

KILL"xxxxxx"
KILL"d:xxxxxx"
KILL"*.BAS" Kills only files with a .BAS suffix

The NAME command

This is actually the RENAME command and takes the form

NAME "d:oldfile" AS "d:newfile"

where d is the drive number, oldfile is the old file name and newfile is the new file name. The drive number must be the same in both statements otherwise a Rename Across Disk error will result.

The COPY command

This command is used to make a copy of a single disk file.

SPECTRAVIDEO

Can only be used with a two drive system.

```
COPY 1 FROM 2
COPY 2 FROM "1:OLD"
```

The Spectravideo COPY command does not allow the filename to be changed while copying.

MSX

```
COPY "d:xxx" TO "d:xxx"  Copy to another disk using a different name
COPY "d:xxx" TO "d:"     Use to copy to another disk using the same name
COPY "d:" TO "d:"        Copy entire disk
```

When used with a single drive system, this command assumes the second drive to be another disk and prompts for that disk to be inserted.

The DSKF command

This command is used to check how much space is left on a disk. It takes the form -

```
DSKF(x)
```

where x is the disk drive number and can be 0, 1 or 2. For MSX, PRINT DSK(1) will display the number of Kbytes left on the disk in drive A. For the Spectravideo it will display the number of 4K blocks left. DSKF(0) and DSKF(1) both refer to drive 1 or A.

The ATTR$ command (SV only)

This command is used to check the ATTRibutes of a disk drive which have been given it by a SET command. It takes the form

```
ATTR$(x)          to check the drive number x
ATTR$(#1)         to check a file which is currently open
ATTR$("d:xxxxxx") to check a file on drive d.

returns R if set to Read after Write
returns P if write protected
```

The SET command (SV only)

This command is used to set the attributes of a disk drive to Read after Write and/or Write Protect. It takes the form

```
SETd,"R"
SETd,"P"
SET"d:xxxxxx","P"
SET"d:xxxxxx","R"
SET#1,"R"
SET#1,"P"
```

where d is the disk drive number, R means Read after Write and P means Write Protect.

Example

```
10 OPEN "1:FILE"FOR OUTPUT AS #1
20 SET#1,"R"
```

PROGRAM DE-BUGGING

This section is designed to assist you to de-bug your own programs and those which you type in but won't run first time.

One of the worst error messages to get is "SYNTAX ERROR in line ....." particularly if that line contains multiple commands separated by a colon (:). Let us look at an example.

```
100 PRINT A$ : PRINT B$ : PRIMT C$
```

Looking at this line, you might immediately spot the incorrect spelling of the third PRINT command. Sometimes you can stare at something for what seems like hours and still the problem does not register.

One trick is to type the whole line again. Hopefully this time your spelling and typing have improved. If the problem persists then the following might help.

List line 100. Move the cursor to the start of the secong PRINT command and insert REM and press ENTER. This means that you have stopped the computer looking at the second and third PRINT commands. Remember that the REM command means "The rest of this line is a remark and contains no commands".

If the program is now RUN, there will be no SYNTAX ERROR in line 100. Now list line 100 again and remove the REM command and insert it in front of the third PRINT command and press ENTER.

RUN the program again and still no SYNTAX ERROR. This means that the error is in the third PRINT command. Now that you only have to examine one word, the error is a little more obvious.

The SYNTAX ERROR may not always be caused by incorrect spelling or even be in the line number specified. For example -

```
10 FOR I=1 TO 10
20 READ A
30 NEXT I
40 DATA A,B,C,D,E,F,G,H,J,I
```

will produce a "Syntax error in 40". Have a look at line 40 - nothing wrong with it.

The actual error is in line 20. READ A tells the computer to expect NUMERIC information in the data line. What the computer found were letters. You would expect this to generate a TYPE MISMATCH error. No, for this program to function correctly, line 20 should be READ A$

Another common error, particularly with games programs where sprites are being built from data statements, is "OUT OF DATA in line ...."

For example

```
10 FOR I=1 TO 10
20 READ A
30 NEXT I
40 DATA 1,2,3,4,5,6,7,8,9
```

will give OUT OF DATA IN 20

The problem is not necessarily in line 20 although this is the first thing to check. If line 20 is correct then the problem lies in the data line being READ by line 20. To explain the program -

Line 10 means repeat all of the lines between here and the NEXT command ten times.
Line 20 looks for the next item in a data line and gives the value to A.
Line 30 means go back to line 10 and check if we have done it 10 times.
Line 40 contains the data which is read by the command in line 20.

The problem lies with line 40. Count the number of data items. There are only 9.

QUESTIONS

SPOT THE ERRORS

1. What is wrong with this?

```
100 PRIMT "ABCD"
```

2. What is wrong with this?

```
100 FOR I=1 TO 5
110 READ A
120 NEXT I
130 DATA A,B,C,D,E
```

3. What is wrong with this?

```
100 FOR I=1 TO 5
110 READ A$
120 NEXT I
130 DATA 1,2,3,4,5
```

4. What is wrong with this?

```
100 FOR I=1 TO 5
110 READ A
120 NEXT I
130 DATA 1,2,3,4,5,6,7,8
```

5. What is wrong with this?

```
100 SCREEN 0
110 LINE(10,10)-(200,10)
```

6. What is wrong with this?

```
100 I=1
110 PRINT I
120 IF I=10 THEN GOTO 140
130 GOTO 100
140 PRINT I
```

7. What is wrong with this?

```
100 SCREEN 0
110 GOTO 140
120 LINE(10,10)-(100,10)
130 GOTO 130
140 SCREEN 2
```

Did we catch you? There is nothing wrong with some of these.

HACKERS CORNER

SUBROUTINE SECTION

Flashing cursor for INPUT

```
10 ON INTERVAL=10 GOSUB 1000 :'vary value for speed
 .
 .
100 A$="":PRINT"PROMPT STRING ?";:INTERVAL ON
110 B$=INKEY$ : IFB$<>CHR$(13)THENA$=A$+B$ :
PRINT B$ : GOTO110
120 INTERVALOFF : PRINT : LOCATE,,0
130 REM continue program using A$
 .
 .
1000 L=1-L : LOCATE,,L : RETURN
```

Ross Moore
Kenilworth, Queensland

GAME LOAD ADDRESSES

	START	FINISH	EXEC.
PINBALL	37123	56000	37123
PYRAMID WARP	40960	49670	40960
POLAR STAR	53100	58000	35860
BATTLESHIP CLAPTON 2	37100	55000	40743

Roger Stockburger
Richmond, NSW

THE SV ROM DECODED

Spectravideo SV-318/328 ROM version 1.1

```
3777  CLS
377E  Screen0 CLS
378B  Screen1 CLS
37B7  Screen2 CLS
37D9  Setup screen as per FE3AH
37E6  Write screen to tape
3831  Read screen from tape
3905  Convert to upper and send to video
390E  Send chr as is
3915  LPTOUT routine
3938  Test printer status
393E  Loc cursor (x in H, y in L)
394D  Chr in A to screen
3985  ESC code lookup
3886  Kill Funct key display
389F  Display function keys
3CC2  Interrupt
3CC9  Check if Modem active
3CE7  Check cassette
3D03  Check keyboard
3DCA  Keypress check
403D  Scan keybd result in A
40B6  Send sound from E
40BE  BEEP
40DD  Delay
```

READERS ADS

Use these pages to sell your surplus equipment, etc. Photocopy this form and insert up to 20 words.. Put it in an envelope with $5 and send to COMPUTER FORUM by the 10th of each month.

.....	.....	.....	.....
.....	.....	.....	.....
.....	.....	.....	.....
.....	.....	.....	.....
.....	.....	.....	.....

FOR SALE. OLIVETTI PRAXIS 35 DAISY WHEEL TYPEWRITER WITH COMPUTER INTERFACE. CAN BE USED WITH ANY CENTRONICS INTERFACE AND ALL MSX COMPUTERS. $290 PLUS $10 FREIGHT ANYWHERE IN AUSTRALIA. ENQUIRE C/o COMPUTER FORUM (09) 339 5087.

WANTED SPECTRAVIDEO 801 DISC CONTROLLER AND A 902 DISC DRIVE. V.A. HOLMES, 106 ARCHIBALD ROAD, KELSTON, AUCKLAND NEW ZEALAND.

MEMORY POINTERS FOR MSX / SPECTRAVIDEO

Label	MSX Addr	SV Addr	Bytes	Explanation
OLDKEY	FBDA	FD75	11	Keyboard matrix
NEWKEY	FBE5	FD80	11	Keyboard matrix
KEYBUF	FBF0	FD8B	40	Keyboard buffer
BOTTOM	FC48	FDE4	2	Bottom of memory pointer
HIMEM	FC4A	FDE6	2	Top of memory pointer
JIFFY	FC9E	FE2E	2	System time clock
CSRSW	FCA9	FA05	1	Cursor switch 1=on, 0=off
CSTYLE	FCAA	FE37	1	Cursor style (Full/half height)
CAPST	FCAB	FE38	1	Caps lock toggle
SCRMOD	FCAF	FE3A	1	Screen mode
DRWFLG	FCBB	FE4A	1	Draw flag
DRWSCL	FCBC	FE4B	1	Draw scale
DRWANG	FCBD	FE4C	1	Draw angle

MSX ROM CALLS

All MSX ROM calls should be made through the BIOS entry table.

MSX1 BIOS ENTRY TABLE

005CH Name: LDIRVM
Function: Move block of main memory to VRAM
Entry: Source in HL
Destination in DE
Length in BC
Returns: None
Modifies: All

005FH Name: CH6MOD
Function: Sets VDP mode as per locn FCAFH (SCRMOD)
Entry: SCRMOD (0 - 3)
Returns: None
Modifies: All

0062H Name: CH6CLR
Function: Change screen colour
Entry: Foreground in FORCLR (F3E9H)
Background in BAKCLR (F3EAH)
Border in BDRCLR (F3EBH)
Returns: None
Modifies: All

0066H Name: NMI
Function: Non-maskable interrupt
Entry: None
Returns: None
Modifies: None

0081H Name: SETMLT
Function: Set VDP for multi colour mode
Entry: MLTNAM,MLTCGP,MLTCOL,MLTATR,MLTPAT
Returns: None
Modifies: All

0084H Name: CALPAT
Function: Get sprite pattern table address
Entry: Sprite ID in A
Returns: Address in HL
Modifies: AF,DE,HL

0087H Name: CALATR
Function: Get sprite attribute table addr.
Entry: Sprite ID in A
Returns: Address in HL
Modifies: AF,DE,HL



RENUM CHECK

Before re-numbering a program (or to check a program for any undefined line numbers), you can do a check as follows — Select a number higher than the last line number, eg 9000, and type RENUM 9000, 9000

This will show up any undefined GOTO, GOSUB, RESTORE, RESUME commands before all but the undefined lines are re-numbered. Much easier to debug.

CAPS LOCK

When keyboard input is required to be in upper case, include the following early in your program —

```
POKE-456,32:OUT136,15:OUT140,255 (SVonly)
```

This enforces upper case and toggles the caps lock LED on. For lower case use —

```
POKE-456,0:OUT136,15:OUT140,223 (SVonly)
```

Our thanks to Ross Moore of Kenilworth for these hints.

COMPUTER KIDS

Hi kids! This month we have a little program that prints a message from Computer Forum. Enter the data lines carefully otherwise you will get the wrong message.

Bye for now, Wendy.

SOLUTION TO WORDSEARCH

T	R	I	H	S	T	C	M
H	F	X	D	N	A	S	A
O	P	L	S	U	N	H	E
N	M	H	U	O	Q	O	R
G	Z	A	R	T	I	R	C
S	K	T	F	U	N	T	N
B	A	T	H	E	R	S	U
G	N	I	M	M	I	W	S

Suitable for SV and MSX

```
10 CLS : WIDTH 40
20 FOR I=1 TO 51 : READ A
30 M$=M$+CHR$(A)
40 NEXT I
50 LOCATE 0,12
60 PRINT LEFT$(M$,40)
70 FOR T=1 TO 50 : NEXT T
80 M$=RIGHT$(M$,50)+LEFT$(M$,1)
90 GOTO 50
100 DATA 77,69,82,82,89,32
110 DATA 67,72,82,73,83,84
120 DATA 77,65,83,32,102,114
130 DATA 111,109,32,67,79,77
140 DATA 80,85,84,69,82,32
150 DATA 70,79,82,85,77,32
160 DATA 77,97,103,97,122
170 DATA 105,110,101,32,46
180 DATA 46,46,32,32,32
```

PROGRAM LISTINGS

Each issue of COMPUTER FORUM will contain program listings for you to type in. When entering these programs it is a good idea to save off to cassette or disk every 10 or 20 lines and then continue. Then if the power goes off or another major disaster occurs, you will not have to type in the whole lot again.

Another check is to verify each save using the CSAVE? command. To use this command, CSAVE your program, then rewind the tape and type CLOAD? and press enter. This command will then compare the program on the tape with the one in memory, to verify that it was saved correctly.

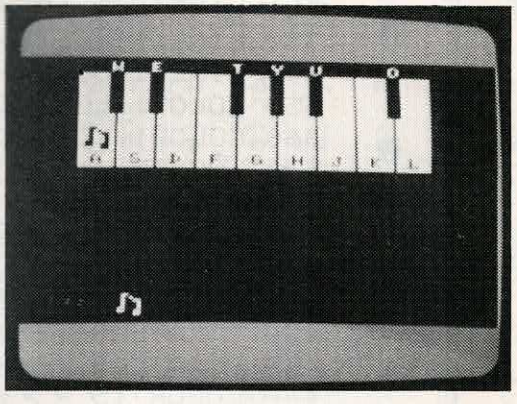
All of the listings are printed in 40 column format. If you enter WIDTH40 before typing them in, you will be able to line up the characters in the listing with the text on screen.

Take care that you do not mistake the letter I for 1 or an O for a zero. Where graphics characters are used in the program, they will be shown as @1@. This symbol will then be repeated at the end of the listing with instructions concerning the graphics character used.

All of the listings have been tested on MSX and SV318/328 computers and do work. Should you have any problems, we suggest that you thoroughly check your work as errors can creep in very easily. Should you not wish to type in these listings, send a blank tape together with $2 to COMPUTER FORUM and we will post a copy to you. Do not forget to tell us which computer you have.

MUSIC

by JAMES POON



This listing is in Spectravideo SV318/328 format. For MSX owners, see the note at the end of the listing.

```
10 REM
20 REM              MUSIC
30 REM
40 REM Introduction.
50 COLOR6,1,1:SCREEN2:FORS=1TO3:PRINT"
 MUSIC ":PRINT:NEXTS
60 FORS=1TO500:NEXTS:SCREEN0
70 REM Set up instructions.
80 COLOR14,1,12
90 PRINTCHR$(27)+"p"
100 LOCATE 10,22:PRINT" MERRY MELODIES "
110 PRINTCHR$(27)+"q"
120 PRINT"           Written by James Poo
n. "
130 FOR S=1 TO 7
140 PRINT
150 NEXT S
160 PRINT" 1.  CAPS. locked!":PRINT
170 PRINT" 2.  Use the keys directed.":P
RINT
180 PRINT" 3.  To change the OCTAVE or P
ITCH of       the note simply use the NU
MBERS        1 to 7 during play.":PRINT:
PRINT
190 PRINT" 4.  PRESS ANY KEY TO START"
   :PRINT
200 IFINKEY$=""GOTO200
210 REM Set up screen display.
220 X=41:Y=50:FILL=1:P$="":OT$="05"
230 PO=20:IU=130:SCREEN1,1
240 LINE(38,10)-(218,80),14,BF
250 LINE(38,10)-(218,80),1,B
260 FOR S=58 TO 198 STEP 20
270 LINE(S,10)-(S,80),1
280 NEXT S
290 LINE(55,10)-(61,40),1,BF
300 LINE(75,10)-(81,40),1,BF
310 LINE(115,10)-(121,40),1,BF
320 LINE(135,10)-(141,40),1,BF
330 LINE(155,10)-(161,40),1,BF
340 LINE(195,10)-(201,40),1,BF
350 COLOR1:LOCATE66,70:PRINT"S"
360 COLOR1:LOCATE86,70:PRINT"D"
370 LOCATE106,70:PRINT"F"
380 LOCATE126,70:PRINT"G"
390 LOCATE146,70:PRINT"H"
400 LOCATE166,70:PRINT"J"
410 LOCATE186,70:PRINT"K"
420 LOCATE206,70:PRINT"L"
430 LOCATE46,70:PRINT"A"
440 COLOR 15
450 LOCATE56,3:PRINT"W"
460 LOCATE76,3:PRINT"E"
```

```
470 LOCATE116,3:PRINT"T"
480 LOCATE136,3:PRINT"Y"
490 LOCATE156,3:PRINT"U"
500 LOCATE196,3:PRINT"O"
510 SOUND6,1:PLAY"sosm300004L8AFAFAFFG"
520 ON INTERVAL= 1 GOSUB 660
530 REM Set up sprites.
540 FORI=1TO8:READA
550 B$=B$+CHR$(A)
560 NEXTI
570 SPRITE$(0)=B$
580 CLICKOFF
590 REM Put sprites in motion.
600 INTERVALON
610 IF FILL> 255 THEN FILL=1
620 FILL=FILL + 1
630 PUT SPRITE 1,(FILL,170),14,0
640 PUT SPRITE 0,(X,Y),1,0
650 GOTO 590
660 INTERVALOFF
670 REM Player input.
680 NT$=INKEY$
690 LOCATE PO,IU:IFPO>247GOTO1230
700 REM Test for invalid input.
710 IF NT$="" THEN NT$=" ":RETURN
720 IF (ASC(NT$)<49) THEN RETURN
730 IF (ASC(NT$)>89) THEN RETURN
740 IF (ASC(NT$)>55) AND (ASC(NT$)<65)
   THEN RETURN
750 IF (ASC(NT$)>65) AND (ASC(NT$)<68)
   THEN RETURN
760 IF (ASC(NT$)>72) AND (ASC(NT$)<74)
   THEN RETURN
770 IF (ASC(NT$)>76) AND (ASC(NT$)<79)
   THEN RETURN
780 IF (ASC(NT$)>79) AND (ASC(NT$)<83)
   THEN RETURN
790 IF (ASC(NT$)>85) AND (ASC(NT$)<87)
   THEN RETURN
800 IF (ASC(NT$)>87) AND (ASC(NT$)<89)
   THEN RETURN
810 IFNT$="A"THENGOSUB1080
820 IFNT$="S"THENGOSUB1090
830 IFNT$="D"THENGOSUB1100
840 IFNT$="F"THENGOSUB1110
850 IFNT$="G"THENGOSUB1120
860 IFNT$="H"THENGOSUB1130
870 IFNT$="J"THENGOSUB1140
880 IFNT$="K"THENGOSUB1150
890 IFNT$="L"THENGOSUB1160
900 IFNT$="W"THENGOSUB1170
910 IFNT$="E"THENGOSUB1180
920 IFNT$="T"THENGOSUB1190
930 IFNT$="Y"THENGOSUB1200
940 IFNT$="U"THENGOSUB1210
950 IFNT$="O"THENGOSUB1220
960 IF NT$="1"THENPLAY"O1":OT$="O2":
   RETURN
970 IF NT$="2"THENPLAY"O2":OT$="O3":
   RETURN
980 IF NT$="3"THENPLAY"O3":OT$="O4":
   RETURN
990 IF NT$="4"THENPLAY"O4":OT$="O5":
   RETURN
1000 IF NT$="5"THENPLAY"O5":OT$="O6":
    RETURN
1010 IF NT$="6"THENPLAY"O6":OT$="O7":
    RETURN
1020 IF NT$="7"THENPLAY"O7":OT$="O8":
    RETURN
1030 PLAY P$
1040 PRINT NT$
1050 RETURN
1060 DATA 48,40,38,34,34,98,102,6
1070 REM Branching for each note played
1080 X=41:Y=50:P$="C7":PO=PO+8:RETURN
1090 X=61:Y=50:P$="D7":PO=PO+8:RETURN103
0
```

```
1100 X=81:Y=50:P$="E7":PO=PO+8:RETURN
1110 X=101:Y=50:P$="F7":PO=PO+8:RETURN
1120 X=121:Y=50:P$="G7":PO=PO+8:RETURN
1130 X=141:Y=50:P$="A7":PO=PO+8:RETURN
1140 X=161:Y=50:P$="B7":PO=PO+8:RETURN
1150 X=181:Y=50:P$="C7":PO=PO+8:GOTO 125
0
1160 X=201:Y=50:P$="D7":PO=PO+8:GOTO 125
0
1170 X=50:Y=41:P$="D-7":PO=PO+8:RETURN10
30
1180 X=70:Y=41:P$="E-7":PO=PO+8:RETURN
1190 X=110:Y=41:P$="G-7":PO=PO+8:RETURN
1200 X=130:Y=41:P$="A-7":PO=PO+8:RETURN
1210 X=150:Y=41:P$="B-7":PO=PO+8:RETURN
1220 X=190:Y=41:P$="D-7":PO=PO+8:GOTO 12
50
1230 REM Branching for clearing text.
1240 COLOR1:LOCATE20,130:PRINTSTRING$(39
,201):PO=20:COLOR15:GOTO680
1250 REM Subroutines for handling
        notes: "O" "K" "L" in the
       next octave
1260 PLAY OT$
1270 IF OT$="02" THEN PLAY P$+"01":RETUR
N 1040
1280 IF OT$="03" THEN PLAY P$+"02":RETUR
N 1040
1290 IF OT$="04" THEN PLAY P$+"03":RETUR
N 1040
1300 IF OT$="05" THEN PLAY P$+"04":RETUR
N 1040
1310 IF OT$="06" THEN PLAY P$+"05":RETUR
N 1040
1320 IF OT$="07" THEN PLAY P$+"06":RETUR
N 1040
1330 IF OT$="08" THEN PLAY P$+"07":RETUR
N 1040
```

NOTE: MSX OWNERS MAKE THE FOLLOWING ALTERATIONS -

ADD LINE 45

```
45 OPEN"GRP:"AS#1
```

DELETE LINE 580

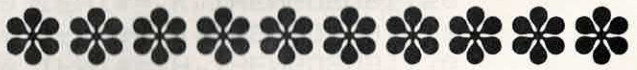
CHANGE LINES 50, 230, 690, 1040, 1240 AS FOLLOWS -

```
50 COLOR6,1,1:SCREEN3:FORS=1TO3:PRESET(0
,S*35+30):PRINT#1,"  MUSIC ":NEXTS
230 PO=20:IU=130:SCREEN2,1,0
690 PRESET( PO,IU):IFPO>247GOTO1230
1040 PRINT#1,NT$
1240 COLOR1:PRESET(20,130):PRINT#1,STRIN
G$(39,219):PO=20:COLOR15:GOTO680
```

LINES 350 TO 430 AND 450 TO 500 - CHANGE ALL LOCATE COMMANDS TO PRESET AND ALL PRINT COMMANDS TO PRINT#1, LIKE THIS

```
350 COLOR1:PRESET(66,70):PRINT#1,"S"
```

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SYNTAX ERRORS etc.

There was an error in the CAPS LAMP FLASH (SV Version) - page 8, Volume one number 4. It should have read -
Change line 10 to read - OUT 140,63
and line 30 to read - OUT 140,31
and press CAPS LOCK before running each time

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